Research Methodology
Course Progress 44.72% (11 of 12)
1_ summary writing
Experimental skills
Literature Survey from Home
Quiz: Week 2 : Assignment
Week 2 Feedback Form: Research Methodology
Week 3 - Data analysis and Modelling skills
Data analysis Part 1
Data analysis Part 2
Modelling skills Part 1
Modelling skills Part 2
Text file of data analysis
Safety in laboratory
Quiz: Week 3 : Assignment
Week 3 Feedback Form: Research Methodology
Week 4: Technical presentation and writing

Assessment

Week 3 : Assignment

The due date for submitting this assignment has passed. Your last recorded submission was on 2026-08-02, 22:54 IST.

Due date on 2026-08-14, 23:59 IST

1. Safety must be the most important discipline of our routine when we are in laboratory or industrial environment. The common sources which have potential to cause damage, available in our lab towards which we need to be careful are- 1 Point

- [x] Sources of heat
- [x] Chemicals (Acids, Bases etc.)
- [x] Pressurized gas cylinders
- [x] Electricity and water source

Yes, the answer is correct. Score: 1

Accepted answer is Sources of heat, Chemicals (Acids, Bases etc.), Pressurized gas cylinders, Electricity and water source

2. Which of the following statistics is used to quantify errors in parameter estimates? 1 Point

- ( ) Mode
- ( ) Skewness
- (•) Standard deviation
- ( ) Median

Yes, the answer is correct. Score: 1

Accepted answer is Standard deviation

3. A temperature sensor is installed at the bottom of the cooling tower storage tank to measure the temperature of outlet water. The sensor did not function properly for some time and gave data with higher temperature values. The conclusion we can draw from the process is- 1 Point

- ( ) Since data with different values were recorded, this increased the possibility of wide analysis thus very useful for our work
- (•) The data obtained cannot be reliable because these are not true data from the system but the malfunctioned one
- ( ) The data can be used as with good accuracy using the predictive type data analysis where we can assume the extent of error and finally subtract the error from final value
- ( ) None of these

Yes, the answer is correct. Score: 1

Accepted answer is The data obtained cannot be reliable because these are not true data from the system but the malfunctioned one

4. Which of the following can be treated as deterministic signals? 1 Point

- ( ) Measurements of atmospheric temperature
- ( ) A sine wave corrupted by a rectangular pulse
- ( ) Estimates of signals from measurements
- (•) None of the above

No, the answer is incorrect. Score: 0

Accepted answer is A sine wave corrupted by a rectangular pulse

5. Identify the incorrect statement(s) among the following w.r.t. non-linear models. 1 Point

- [x] Linearization involves mere discarding of the non-linear terms in a model.
- [ ] The standard Taylor's series-based linearized model changes with the nominal point of linearization.
- [x] A non-linear model can never yield a zero-valued output to a non-zero input.
- [x] Estimates of linear models have smaller errors than those of non-linear models.

Yes, the answer is correct. Score: 1

Accepted answer is Linearization involves mere discarding of the non-linear terms in a model., A non-linear model can never yield a zero-valued output to a non-zero input., Estimates of linear models have smaller errors than those of non-linear models.

6. Which of the following transforms are used to detect periodic (oscillatory) components in a signal? 1 Point

- ( ) Hilbert transform
- ( ) z-transform
- (•) Fourier transform
- ( ) All of the above

Yes, the answer is correct. Score: 1

Accepted answer is Fourier transform

7. The ratio of errors in parameter estimates of a model from two data sets with SNR 9 and 100 is 1 Point

- ( ) 3 : 10

- 5 : 1
- 10 : 3
- 10 : 0.9

Yes, the answer is correct. Score: 1 | Accepted answer is 10 : 3

8. Which of the following is true concerning a parameter estimate? 1 Point

- It is in general close to the true value with high probability.
- When averaged across a million experimental records under same operating conditions, the estimate equals the true value.
- Least squares method always gives the estimate with lowest error.
- It is a mathematical function of the observations.

Yes, the answer is correct. Score: 1 | Accepted answer is It is a mathematical function of the observations.

9. Which of the following statements is true? 1 Point

- [ ] Deterministic phenomena produce outcomes that can be predicted accurately.
- [ ] Random processes occur randomly.
- [x] A process can produce a mix of deterministic and stochastic effects.
- [ ] Signals that are random can never take negative values.

Partially Correct. Score: 0.5 | Accepted answer is Deterministic phenomena produce outcomes that can be predicted accurately., A process can produce a mix of deterministic and stochastic effects.

10. An estimator is said to be most efficient when 1 Point

- The parameters can be estimated in the least possible time.
- The estimator provides accurate estimates with the smallest number of observations.
- Standard error in the parameter estimates is lowest when compared to that of all other estimators.
- It estimates parameters accurately even in presence of outliers.

Yes, the answer is correct. Score: 1 | Accepted answer is Standard error in the parameter estimates is lowest when compared to that of all other estimators.